

sense of delight and jubilation. Yet, her letter also sounded a note of bewilderment and reservation: she did not know of any experience or story like her own and was perplexed to find, in all she had read, that the achievement of stereoscopy in adult life was “impossible.”

Indeed, what Sue described to me in her letter went completely against the current dogma of “critical periods” in sensory development—the notion that stereoscopy (like many other aspects of visual perception, and like language, as well)—had to be acquired in the first three or four years of life, or it could never be acquired, for the critical brain cells and circuitry needed for stereovision would fail to develop.

Long suspected by surgeons operating on children with strabismus, this notion of a critical period seemed to be confirmed by the famous experimental work of David Hubel and Torsten Wiesel, who showed that if kittens were rendered strabismic by detaching an eye muscle, binocular depth cells would fail to develop in their brains, and they would lack stereovision. It was only when Sue learned of these experiments—she was a college student at the time—that she realized she herself might be stereoblind, like the kittens. This, indeed, is the vivid opening scene in her narrative:

Stereoblind? Was I stereoblind? I looked around. The classroom didn't seem entirely flat to me. I knew that the student sitting in front of me was located between me and the blackboard because the student blocked my view of the blackboard. When I looked outside the classroom window, I knew which trees were located further away because they looked smaller than the closer ones. The footpath outside the window appeared to narrow as it extended out into the distance. Through cues like these, I could judge depth and distance. I knew the world was in 3D. Yet, my professor implied that there was another, different way to see space and depth. He called this way of seeing stereopsis. I couldn't imagine what he was talking about.

When Sue next went to her eye doctor for a routine check, she asked him to check whether she had stereovision. He brought out a stereoscope and test stereo pictures. Sue could not “get” any of them, could not imagine what “getting” them would be like. Would it be possible for her to gain stereovision, she asked? The doctor replied, no, it was much too late, and added, “Stereopsis is just a little fine-tuning for the visual system. You don’t need stereovision because you don’t have stereovision.”

Sue accepted that she would never have stereovision and got on with her life, becoming a teacher and researcher, marrying, and raising a family. Somewhere, at the back of her mind—for she is a scientist and incessantly curious about how the world works—was a question: what could stereovision be like? And yet, her life, visually and otherwise, was full and rich, and she did not “miss” stereo or think of it too much. So thirty years later, when she finally sought vision therapy and unexpectedly gained stereovision, this came as a bonus, a miraculous complement to her other visual improvements.

Sue exulted in her newfound sense of stereoscopic depth. She found it much more than “fine-tuning”—it was an entirely new way of seeing. “People who have always had stereopsis,” she said to me, “take it for granted. They have no idea how wonderful it is. You have to have been stereoblind for half a century and then acquire it to value it properly.”

How was Sue able to acquire, essentially, a whole new sense so long after the “critical period”? I was as puzzled by Sue’s story as she was. And I was intrigued, for I myself had never taken stereoscopy for granted. On the contrary, I was something of a stereophile, having played with 3D drawings and Victorian stereo viewers as a child and later experimented with stereo photography. So I arranged to meet Sue again, and this inspired me to write an article about her experiences in 2006.^{[1](#)}

But that was not the end of the matter.